

Abstract No.

LC-MS/MS-Based Metabolomic Analysis of ATRA and Curcumin Combination Treatment in U87-MG Glioblastoma Cells

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Glioblastoma (GBM) is an aggressive brain tumor characterized by metabolic plasticity and resistance to conventional therapies. Understanding treatment-induced metabolic alterations may provide insight into mechanisms of therapeutic response and potential metabolic vulnerabilities in GBM. This study aimed to characterize metabolic alterations induced by all-trans retinoic acid (ATRA), curcumin (CURC), and their combination in U87-MG glioblastoma cells using LC-MS/MS-based untargeted metabolomics. U87-MG cells were treated with ATRA (75 μ M), CURC (20 μ M), or their combination for 72 h. Intracellular metabolites were analyzed using UHPLC coupled to an Orbitrap Exploris 240 high resolution mass spectrometer. Data processing and metabolite annotation were performed using Compound Discoverer 3.3. Significantly altered metabolites ($p < 0.05$ and $|\log_2 FC| > 1$) were subjected to pathway enrichment analysis using MetaboAnalyst 6.0 and KEGG. Metabolomic profiling identified 46, 77, and 74 significantly altered metabolites following ATRA, CURC, and combination treatment, respectively. ATRA affected sphingolipid, tryptophan, and glutathione metabolism, whereas CURC predominantly altered methionine and purine metabolism. Combination treatment induced broader metabolic changes involving spermidine biosynthesis, methionine and purine metabolism, glutathione metabolism, pantothenate and CoA biosynthesis, betaine metabolism, thiamine metabolism, and sphingolipid metabolism. ATRA and CURC induced distinct metabolic responses in glioblastoma cells, while their combination was associated with broader metabolic reprogramming involving amino acid, nucleotide, lipid, polyamine, and redox-related pathways. These findings provide insight into treatment-associated metabolic adaptations in GBM cells and the molecular mechanisms underlying the response of glioblastoma cells to combined ATRA and CURC treatment. Further studies integrating metabolomic findings with functional validation and additional molecular approaches are necessary to clarify the biological significance of these metabolic alterations and their potential relevance to future therapeutic strategies for GBM.

Keywords: Glioblastoma; ATRA; Curcumin; Untargeted Metabolomics; LC-MS/MS; Metabolic Reprogramming.



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